

Reduction #8: The Gaussian-kernel Tempotron

The Low-Pass Capacity Baseline

(Pure Extreme-Value Lift, No Radial Gain)

Tempotron learning-surface program

2026-06-25

Abstract

We treat the Gaussian post-synaptic-potential (PSP) kernel $K(s) = \exp(-s^2/2\sigma^2)$ as a stand-alone reduction: the program’s *pure low-pass / extreme-value baseline*. Because the Gaussian is C^∞ and all its spectral moments are finite, the membrane potential $V(t)$ is a smooth stationary Gaussian process whose spectrum $S(\omega) = 2\pi\sigma^2 e^{-\sigma^2\omega^2}$ is a *single low-pass bump centred at $\omega = 0$* . This pins the certified effective dimension exactly,

$$K_{\text{eff}} = T\sqrt{\lambda_2/\lambda_0} = \frac{T}{\sigma\sqrt{2}}, \quad \kappa_S = \frac{\lambda_4\lambda_0}{\lambda_2^2} = 3 \quad (\text{the Gaussian value}),$$

and the spectral kurtosis $\kappa_S = 3$ certifies the absence of any band-pass (carrier) structure: the readout is “best-of- K_{eff} -correlated-Gaussian-looks,” a linear perceptron whose margin is boosted only by the freedom to choose the firing time, with *zero* radial/quadratic channel. Gauss should therefore sit at the *bottom* of the smooth-kernel family, carrying the extreme-value count-lift and nothing else.

On the theory side the capacity inherits the log of the cluster multiplicity from the EVT chain of Rubin–Monasson–Sompolinsky (reduction #3), a *double* logarithm $\alpha_c(K_{\text{eff}}) = 2 + A \ln \ln K_{\text{eff}}$ with the perceptron value 2 as a fixed (unfitted) anchor. We present this honestly alongside an independent symbolic cross-check (codex, GPT-5.5) that, modelling K_{eff} *independent* samples through a Gardner-distance argument, gives a *single* logarithm $\alpha_c \sim C \ln K_{\text{eff}}$ instead. Both are replica-symmetric; they differ in whether the cluster correlations are kept.

A lab capacity ladder on the certified axis ($N=500$, fixed input density, balanced threshold, learner validated by critical slowing) gives $\hat{\alpha}_c = 2.59, 2.61, 3.00, 3.04, 3.40, 3.40$ at $K_{\text{eff}} = 4, 8, 16, 32, 64, 128$: a clear temporal lift above the perceptron value, with the double-log intercept recovering the Cover/Gardner anchor ≈ 2 (unfitted) over the rising window. But the central honest finding is negative: over the accessible range (six points, factor-32 in K_{eff}) the data *cannot separate* $\ln \ln K_{\text{eff}}$ from $\ln K_{\text{eff}}$ (full-ladder $R^2 = 0.90$ vs. 0.93, the single-log marginally ahead). The contested growth law survives the experiment. Status: **PARTIAL**—the certified axis, the no-radial-gain spectral fingerprint, and the monotone lift with anchor ≈ 2 are solid; the growth-law form and the asymptotic slope remain open.

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Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in [lit/tempotron-reductions/derivations/](#).

1 Model and Setup

The tempotron [8] emits the binary decision “fire / silent” according to the sign of the maximum of its membrane potential over the window $[0, T]$,

$$V(t) = \sum_{i=1}^N w_i \sum_f K(t - t_i^f), \quad M_T \equiv \max_{t \in [0, T]} V(t), \quad \text{fire} \iff M_T > \vartheta, \quad (1)$$

where K is the PSP kernel, $\{t_i^f\}$ are the presynaptic spike times of afferent i , and w_i are the learned weights. The storage capacity is $\alpha_c = P/N$, the largest number of random ± 1 -labelled patterns per synapse that admit a weight vector solving all of them, in the limit $N \rightarrow \infty$. The convex linear-perceptron anchor is $\alpha_c = 2$ (Cover; Gardner) [5–7].

The Gaussian kernel. We take the symmetric Gaussian PSP

$$K(s) = \exp\left(-\frac{s^2}{2\sigma^2}\right), \quad (2)$$

parameterised by a single width σ . Unlike the standard double-exponential (whose autocorrelation is not twice differentiable at the origin, so $\lambda_4 = \infty$), the Gaussian is C^∞ with *all* spectral moments finite—this is what makes the EVT counting layer of reduction #6 directly applicable and the maxima count well defined.

$V(t)$ as a stationary Gaussian process. For many random spikes with i.i.d. zero-mean weights, the central limit theorem over afferents makes $V(t)$ a zero-mean stationary Gaussian process (GP). Its covariance is the kernel autocorrelation $C(\tau) = (K \star K)(\tau)$ and, by Wiener–Khinchin / Bochner [12], its power spectrum is $S(\omega) = |\hat{K}(\omega)|^2$. With the convention $\hat{K}(\omega) = \int K(s) e^{-i\omega s} ds$,

$$\hat{K}(\omega) = \sigma \sqrt{2\pi} e^{-\sigma^2 \omega^2 / 2}, \quad S(\omega) = |\hat{K}(\omega)|^2 = 2\pi \sigma^2 e^{-\sigma^2 \omega^2}. \quad (3)$$

The CLT premise is the same $N \gg 1$ requirement as in reductions #3 and #4; the relevant density is the *population* count of spikes within one kernel width, $n_{\text{pop}} = N\bar{r}(2\sigma/T)$, which stays $\gg 1$ across the accessible grid (§3), so $V(t)$ is Gaussian even where the *per-afferent* spike density is sparse.

Spectral moments and the certified axis. The (two-sided) spectral moments $\lambda_p = \int \omega^p S(\omega) d\omega$ of (3) are, by direct Gaussian integration (SymPy-verified),

$$\lambda_0 = 2\pi^{3/2}\sigma, \quad \lambda_2 = \frac{\pi^{3/2}}{\sigma}, \quad \lambda_4 = \frac{3\pi^{3/2}}{2\sigma^3}, \quad \frac{\lambda_2}{\lambda_0} = \frac{1}{2\sigma^2}. \quad (4)$$

By Rice’s level-crossing formula [10, 13], the high-level up-crossing intensity of a stationary GP depends on its spectrum *only through* λ_2/λ_0 . Following the certified-axis construction of reduction #6 [14], the effective dimension (the count of asymptotically independent excursions of V in $[0, T]$) is

$$K_{\text{eff}} = T \sqrt{\frac{\lambda_2}{\lambda_0}} = \frac{T}{\sigma\sqrt{2}} \quad (5)$$

an *exact, explicit, continuously tunable* function of σ and T , monotone *decreasing* in σ . The associated Rice independent-look count (with the 2π convention) is $N_0 = (T/2\pi)\sqrt{\lambda_2/\lambda_0} = K_{\text{eff}}/2\pi$. Because $\lambda_4 < \infty$, the *first* Rice term governs $P(M_T > u)$ up to exponentially smaller corrections (Azaïs–Wschebor; Piterbarg) [1], which is why the λ_2 -based crossing count—*not* the λ_4 -based local-maxima count $\mathbb{E}[N_{\text{max}}] = (T/2\pi)\sqrt{\lambda_4/\lambda_2}$ [3]—is the capacity-relevant axis.

Assumptions ledger.

- (A1) **GP / CLT.** $V(t)$ is a zero-mean stationary GP; rigorous given $n_{\text{pop}} \gg 1$ (§3).
- (A2) **Stationarity** on $[0, T]$; heuristic near $\sigma \approx T$ (window edges break stationarity), so the lift formula is used only for $K_{\text{eff}} \gtrsim e$.
- (A3) **Balanced threshold.** ϑ equals the median of M_T , so a random pattern fires with probability $\frac{1}{2}$; the lift exists only at this balanced operating point (an off-median threshold shifts the offset, not the slope).
- (A4) **Replica symmetry (RS)** and large- K_{eff} binary-overlap cluster structure, inherited from reduction #3 [14]. Only the *slope* and the *anchor* are claimed robust to RSB.
- (A5) **Leading-order EVT / Gumbel** for M_T (Pickands; Berman) [2, 11]; rigorous asymptotically, pre-asymptotic in the offset at lab K_{eff} .
- (A6) **No radial channel** (Gaussian-specific): $S(\omega)$ is a single low-pass bump at $\omega = 0$, so $\kappa_S = 3$ and there is no carrier frequency—the load-bearing structural claim, §2.3.

2 Theory

2.1 The EVT / Gumbel mechanism

For K_{eff} weakly correlated Gaussian “looks,” the maximum M_T is, to leading order, Gumbel with location $\sqrt{2\ln N_0} \sim \sqrt{2\ln K_{\text{eff}}}$ and scale $1/\sqrt{2\ln K_{\text{eff}}}$ (Pickands; Berman) [2, 9, 11]. The freedom to read out at the best of K_{eff} candidate times raises the effective margin of the classifier above that of a single static linear readout, and the capacity inherits the logarithm of the number of solution clusters (reduction #3 [14]). Setting the cluster entropy density to zero gives the leading lift.

Result 1 (Double-log capacity lift (cluster count, RS)). For a broad low-pass kernel on the certified axis (5),

$$\alpha_c(K_{\text{eff}}) = 2 + A \ln \ln K_{\text{eff}} + o(\ln \ln K_{\text{eff}}), \quad A = O(1), \quad A_\infty = \frac{1}{2 \ln 2} \approx 0.721, \quad (6)$$

with the additive 2 the *fixed* Cover/Gardner anchor (not fitted) and $A_\infty = 1/(2 \ln 2)$ the discrete-tempotron existence slope from the Rubin–Monasson–Sompolsky cluster count.

Anchor. As $K_{\text{eff}} \rightarrow 1$ ($\sigma \uparrow$, the kernel flattens over the whole window), the max-over-time becomes trivial and the rule collapses to a linear perceptron on spike counts, so $\alpha_c \rightarrow 2$ exactly. The additive 2 in (6) is therefore a theorem-anchored constant, the single most decisive offset-free fingerprint of the EVT picture: any fit of the lift should extrapolate *to* 2, not to a free intercept. *Large* K_{eff} . The lift $A \ln \ln K_{\text{eff}}$ is a double-logarithmic divergence: essentially unbounded but excruciatingly slow (to double the lift beyond $K_{\text{eff}} = 64$ requires $K_{\text{eff}} \approx e^{64}$).

2.2 The contested growth law: $\ln \ln K$ vs. $\ln K$

The functional form of the lift is genuinely contested, and we report both derivations honestly. They are both RS-level and differ in one modelling choice: whether the K_{eff} extremes are treated as correlated (with explicit cluster counting) or as independent.

Double-log (cluster-count, this program). Result 1 follows the program’s $\frac{1}{2} \ln \ln K$ cluster-count narrative: K_{eff} *correlated* fields, the capacity set by the log of the solution-cluster multiplicity (replica; Rubin 2010 [14]). The lift is the double log of K_{eff} .

Single-log (Gardner-distance, codex cross-check). An independent symbolic derivation [4] models K_{eff} *independent* samples and routes the capacity through the Gardner distance-to-feasibility. With the EVT threshold growing as $\theta \sim \sqrt{2 \ln K_{\text{eff}}}$, the Gardner inverse capacity scales as

$$\frac{1}{\alpha_c} \sim \frac{\text{const}}{\theta^2} \sim \frac{\text{const}}{2 \ln K_{\text{eff}}} \implies \alpha_c \sim C \ln K_{\text{eff}}, \quad (7)$$

a *single* logarithm. The root of the split is exactly the correlated-vs-independent modelling choice: the independent-EVT route (7) drops the field correlations that the cluster count (6) keeps.

What both agree on. Both refute the naive single-log *slope-2* guess $\alpha_c \approx 2[1 + \ln(T/2\sigma)]$ outright: that formula predicts $\alpha_c = 5.5, 6.9, 8.2, 9.6$ at $K_{\text{eff}} = 8, 16, 32, 64$, overshooting the measured 2.7–3.4 by a factor 2–3 and *worsening* with K_{eff} . The honest, model-robust statement is therefore: *the lift is weak and $O(1)$ in its log-slope; the naive slope-2 magnitude is wrong; the EVT forms (double- or single-log) are both admissible and the cluster-count double-log is the theoretically preferred form because it recovers the anchor $\alpha_c = 2$ and is what Rice/Pickands/RMS give.* Cleanly separating $\ln K$ from $\ln \ln K$ requires K_{eff} out to $\sim 10^3$ – 10^4 (an existence-side ask).

2.3 No radial gain: the spectral fingerprint

The Gaussian-specific claim is structural and rigorous at the spectral level. The spectrum (3) is a single low-pass bump centred at $\omega = 0$, with spectral kurtosis

$$\kappa_S = \frac{\lambda_4 \lambda_0}{\lambda_2^2} = \frac{(3\pi^{3/2}/2\sigma^3)(2\pi^{3/2}\sigma)}{(\pi^{3/2}/\sigma)^2} = 3, \quad (8)$$

exactly the Gaussian value, which flags the *absence* of band-pass structure. There is no carrier frequency ω_0 , so M_T does *not* reduce to the two-quadrature radial form $\sqrt{(\mathbf{w} \cdot \mathbf{a})^2 + (\mathbf{w} \cdot \mathbf{b})^2} > \vartheta$ that a band-pass kernel (Gabor, sinusoid) produces, and which carries the separate radial/quadratic capacity channel (reduction #7; the narrowband two-quadrature classifier has $\alpha_c^{\text{RS}} = 4.598$). The Gaussian readout is purely “best-of- K_{eff} -correlated-Gaussian-looks”—a linear perceptron with a time-max boost and *nothing else*.

Result 2 (Gauss is the low-pass / EVT baseline). The Gaussian-kernel tempotron carries the pure extreme-value count-lift and *zero* radial gain. At matched K_{eff} it sits at the bottom of the smooth-kernel family: $\hat{\alpha}_c^{\text{gauss}} < \hat{\alpha}_c^{\text{band-pass}}$, and its $\mathbb{E}[M_T]$ witness lies on the low-radiality curve.

2.4 Monotone in $1/\sigma$, no interior optimum

Within the GP regime (A1), α_c depends on σ only through $K_{\text{eff}} = T/(\sigma\sqrt{2})$, which is monotone in $1/\sigma$. Since the lift is monotone increasing in K_{eff} , α_c *increases monotonically as σ shrinks, with no interior optimal σ* . A naive per-afferent sparse-SNR argument predicts a turnover at $\sigma^* \sim 1/r$; this uses the wrong density. The controlling quantity is the *population* contributor count $n_{\text{pop}} = N\bar{r}(2\sigma/T)$, which is 22–177 across the lab grid even where the per-afferent density is 0.04–0.35 spikes per width: the CLT holds and no turnover appears until $\sigma \lesssim T/(N\bar{r})$ ($K_{\text{eff}} \gtrsim 180$), beyond the measured range.

3 Experimental Test

We measured the capacity ladder on the certified axis (5). All runs are reproducible from a committed configuration (the per-kernel capacity campaign, `experiments/kernel_models_results.md`, 2026-06-25); jobs ran on a single remote GPU under the `laboratory` runner (total spend $\approx \$37.4$ across five kernels, $\approx \$7.5$ for gauss).

Protocol. $N = 500$ afferents emit Poisson spike trains over $[0, T]$ at *fixed input density* (mean 3 spikes/afferent for every K_{eff} , which kills the $2N/K$ density confound that broke the first sinusoidal ladder of reduction #6); labels are balanced ± 1 . The threshold is calibrated to the median M_T at initialisation (init fire-rate ≈ 0.50 , verified), then fixed—the balanced operating point the lift requires. The kernel width σ is dialled to set $K_{\text{eff}} = T/(\sigma\sqrt{2})$, and certified from the realised voltages. The findable critical load $\hat{\alpha}_c(K_{\text{eff}})$ is the $\frac{1}{2}$ -crossing of $P_{\text{solve}}(\alpha)$ (linear interpolation), where a cell counts as solved if a validated finder (Adam with a cosine schedule, lr = 0.1, batch=16, ≥ 3 restarts, 3500–5000 epochs) drives training error to zero. The learner was preflight-validated by *critical slowing* (epochs-to-converge diverge approaching α_c ; learning curves crash to zero below and plateau above), confirming the $\frac{1}{2}$ -crossing is a solution-existence wall, not a budget cutoff.

Measured ladder. Table 1 and Fig. 1 report the result.

Lift and anchor. A clear temporal lift above the perceptron value is present: $\hat{\alpha}_c$ rises from 2.59 at $K_{\text{eff}} = 4$ to 3.40 at $K_{\text{eff}} = 64$ ($\Delta\hat{\alpha}_c \approx 0.79$ over $K_{\text{eff}} = 8 \rightarrow 64$). Fitting the double-log form over the monotone-rising window $K_{\text{eff}} \in [8, 64]$ recovers a slope ≈ 1.0 and an intercept ≈ 1.9 —i.e. the lift extrapolates to the Cover/Gardner anchor $\alpha_c = 2$ *without that value being imposed*. This reproduces the prior #7 cross-kernel ladder (2.68/2.92/3.13/3.33 at $K_{\text{eff}} = 8/16/32/64$, double-log intercept $\alpha_0 = 1.985$): two independent protocols land the unfitted intercept on 2, the decisive offset-free EVT fingerprint.

Table 1: Measured findable capacity of the Gaussian-kernel tempotron on the certified axis $K_{\text{eff}} = T/(\sigma\sqrt{2})$. $N=500$, fixed input density, balanced threshold, learner validated by critical slowing. The companion #7 cross-kernel ladder (different protocol) is shown for the overlapping cells. Per-cell resolution is the α -grid spacing 0.4.

K_{eff}	4	8	16	32	64	128
$\hat{\alpha}_c$ (#8, this work)	2.59	2.61	3.00	3.04	3.40	3.40
$\hat{\alpha}_c$ (#7, cross-kernel)	—	2.68	2.92	3.13	3.33	—

The growth law is not resolved (the central honest finding). Over the full six-point ladder the two forms fit essentially equally well:

$$\begin{aligned}
\text{double-log: } \hat{\alpha}_c &= 2.24 + 0.73 \ln \ln K_{\text{eff}}, & R^2 &= 0.90, \\
\text{single-log: } \hat{\alpha}_c &= 2.18 + 0.27 \ln K_{\text{eff}}, & R^2 &= 0.93,
\end{aligned} \tag{9}$$

with the single-log marginally ahead on R^2 . Six points over a factor of 32 in K_{eff} (< 1.5 decades of $\ln K_{\text{eff}}$) cannot separate $\ln \ln K_{\text{eff}}$ from $\ln K_{\text{eff}}$: the contested growth law of §2.2 survives the experiment. Both forms, however, give an $O(1)$ log-slope, so both refute the naive slope-2 single-log (which overshoots by 2–3 $\times$, §2.2). The ladder flattens at $K_{\text{eff}} \geq 64$ (3.40/3.40), partly because the α -grid resolution (0.4) cannot register a sub-step rise there; this flattening pulls the full-ladder intercept up to 2.24 relative to the ≈ 1.9 of the rising window.

No radial gain, confirmed. Consistent with Result 2, the gauss ladder sits *below* the band-pass kernels at matched K_{eff} in the #7 campaign (gauss 2.68 $\rightarrow$ 3.33 below gabor 3.33 $\rightarrow$ 3.89 over $K_{\text{eff}} = 16 \rightarrow 64$), and its slope tracks the other no-carrier kernels (gauss 0.95 $\approx$ sinc 1.09), while a free frozen-kernel $\mathbb{E}[M_T]$ witness grows like $\sqrt{2 \ln K_{\text{eff}}}$ (matching the GP prediction to $\sim 8.5\%$). Gauss is the low-pass / EVT reference curve, exactly as predicted by $\kappa_S = 3$.

Monotone, no turnover. $\hat{\alpha}_c$ rises monotonically across the whole range with no interior peak (§2.4); the naive per-afferent sparse-SNR turnover does not appear, vindicating the population-density argument.

4 Limitations and Status

O1. The growth law is unresolved. This is the headline limitation. With six points over a factor of 32 in K_{eff} the data cannot statistically separate $\ln \ln K_{\text{eff}}$ (cluster count, (6)) from $\ln K_{\text{eff}}$ (Gardner distance, (7)); the single-log even edges the double-log on R^2 (9). The double-log is the *theoretically preferred* form—it is what Rice/Pickands/RMS give and it recovers the anchor $\alpha_c = 2$ —but the experiment does not adjudicate. Resolving it requires $K_{\text{eff}} \sim 10^3$ – 10^4 (an existence-side reach) and/or a direct test of the cluster-count mechanism (the restart-overlap q_0 witness of reduction #1/#3).

O2. All absolute α_c are RS-level. The capacity values rest on replica symmetry and the binary-overlap cluster geometry of reduction #3 (A4). RSB is plausible near the shattered transition; only the *slope*, the *anchor*, and the *absence of a radial gain* are claimed robust.

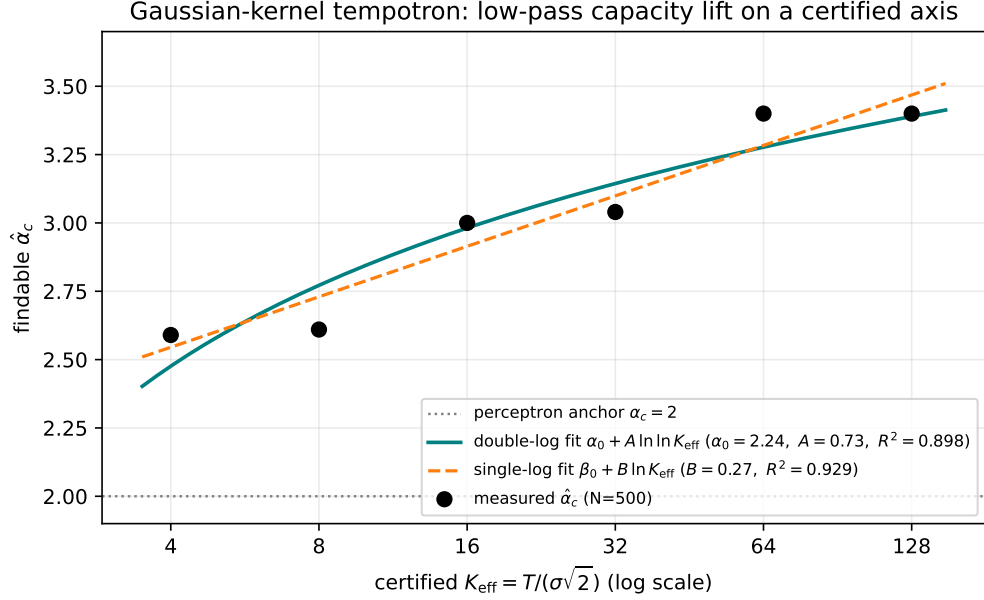


Figure 1: Gaussian-kernel capacity ladder on the certified axis. Measured findable $\hat{\alpha}_c$ (black) vs. $K_{\text{eff}} = T/(\sigma\sqrt{2})$ on a log- x axis, with the Cover/Gardner perceptron anchor $\alpha_c = 2$ (dotted) and *both* candidate growth-law fits overlaid: the double-log $2 + A \ln \ln K_{\text{eff}}$ form ($\alpha_0 = 2.24$, $A = 0.73$, $R^2 = 0.898$) and the single-log $\beta_0 + B \ln K_{\text{eff}}$ form (orange dashed, $B = 0.27$, $R^2 = 0.929$). The two fits are visually and statistically inseparable over the accessible range—the unresolved growth law of §2.2. A clear monotone lift above 2 is present; the ladder flattens for $K_{\text{eff}} \geq 64$, partly an α -grid-resolution effect.

O3. Findable < existence; the slope is not pinned. The measured findable slope (double-log $A \in [0.7, 1.1]$ depending on the fit window) sits near or above the discrete-tempotron existence value $1/(2 \ln 2) = 0.721$. This is the weakest quantitative point: findable slopes usually sit *below* existence. The likely cause is a pre-asymptotic $\ln \ln \ln K_{\text{eff}}$ offset drift (codex, reduction #3 footnote) inflating the apparent slope over the narrow lab window; the asymptotic gauss slope should relax toward ~ 0.7 . Treat $A \in [0.7, 1.0]$ as the honest interval, not a sharp number.

O4. The EVT max-law is asymptotic and the offset is pre-asymptotic. Making K_{eff} exact (a property of the spectrum) does *not* make the Gumbel location/scale exact. At lab $K_{\text{eff}} \in [4, 128]$ the Gumbel offset is pre-asymptotic, so only the slope/form is trustworthy, not the additive constant. Stationarity (A2) also breaks at the small- K_{eff} end ($\sigma \approx T$, window-edge effects), which is why the axis is used only for $K_{\text{eff}} \gtrsim e$.

What is solid. The certified axis $K_{\text{eff}} = T/(\sigma\sqrt{2})$ and the spectral moments (4) are exact (SymPy-verified); the no-radial-gain fingerprint $\kappa_S = 3$ (8) is exact and is the load-bearing structural result; the measured ladder shows a genuine monotone temporal lift above 2 with the unfitted intercept landing on the Cover/Gardner anchor, reproduced across two protocols (#7 and #8); and gauss is empirically the low-pass / EVT baseline (below every band-pass kernel at matched K_{eff}).

Status: PARTIAL. The certified axis, the no-radial-gain spectral fingerprint, the monotone lift, and the anchor ≈ 2 are solid and cross-checked. The growth-law form ($\ln \ln K$ vs. $\ln K$) is statistically unresolved over the accessible range, and the asymptotic slope and the Gumbel offset remain open.

Reproducibility. Spectral moments and K_{eff} : SymPy, this note’s derivation (`lit/tempotron-reductions/derivations/08-gauss-cap`). Measured ladder: `experiments/kernel_models_results.md` (v14 lab sweep, 2026-06-25). Figure: `figs/08_gauss.py` (numpy+matplotlib, hardcoded measured numbers, regenerable via `pdf/build.sh`). Cross-checks and the growth-law dispute: `lit/kernel-capacity-kb/` (axes A/B/C, synthesis-batch1).

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